

PET and Gestational Hypertension 06: Severe Features and the Maternal Safety System

Companion reading handout for simultaneous listening.
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Opening Case

Let us begin with the room changing.

A patient with known gestational hypertension is now thirty four weeks. Her blood pressure is one hundred seventy two over one hundred fourteen. She says the lights look strange. Her platelets are ninety thousand. The fetus is still tracing, but the room no longer feels like routine surveillance.

This is the moment when the clinical frame must change.

You are no longer managing a label. You are preventing the next maternal catastrophe.

Severe features tell you that the endothelial disease is landing in organs that can fail quickly. Severe blood pressure threatens the brain. Low platelets signal microangiopathy. Rising creatinine signals renal injury. Liver enzymes and right upper quadrant pain signal hepatic involvement. Pulmonary edema signals capillary leak and cardiopulmonary danger. Headache and visual symptoms signal cerebral irritability. Eclampsia and HELLP syndrome are not warning signs; they are complications.

The list of severe features is short enough to memorize, but the physiology is what makes it stick. Severe BP is a pressure injury problem. Platelets below one hundred thousand are a microvascular consumption problem. Creatinine above one point one, or doubling of baseline, is a glomerular endotheliosis problem. Liver enzymes at least twice normal are hepatic endothelial and parenchymal injury. Pulmonary edema is capillary leak plus cardiac and fluid-management vulnerability. Persistent neurologic symptoms are cerebral endothelial dysfunction, edema, vasospasm, or hemorrhage until proven otherwise.

This is why the severe-feature list is not arbitrary. It is a map of maternal organs under endothelial attack.

Stroke Prevention

The first emergency target is stroke prevention.

Severe blood pressure means systolic at least one hundred sixty, diastolic at least one hundred ten, or both. If it persists for about fifteen minutes, treat. ACOG recommends treatment as soon as reasonably possible, ideally within thirty to sixty minutes. ISSHP recommends urgent treatment in a monitored setting. The Hadassah protocol uses the same severe threshold.

Do not wait four hours. The four-hour rule belongs to non-severe diagnostic confirmation. Severe BP belongs to stroke prevention.

The target is controlled reduction, not normal pressure.

If you drop the BP too far too fast, you may reduce uteroplacental perfusion. Most protocols aim to bring the patient out of the severe range, often toward about one hundred forty to one hundred fifty five systolic and ninety to one hundred five diastolic, depending on the local protocol.

This is an important distinction at the bedside. You are not treating a chronic outpatient target. You are treating a hypertensive emergency threshold. The immediate win is getting the patient below the range associated with intracranial hemorrhage and stroke, while maintaining maternal and placental perfusion.

Seizure Prevention

The second emergency target is seizure prevention.

Magnesium sulfate is the drug of choice for eclampsia and for seizure prophylaxis in pre-eclampsia with severe features. It is not an antihypertensive. It does not fix the BP. It reduces neuromuscular and central nervous system excitability and lowers the risk of seizures.

The Magpie trial showed that magnesium reduced eclampsia by more than half in women with pre-eclampsia. That is why magnesium is foundational in severe disease. In pre-eclampsia without severe features, use is more controversial and often local-protocol dependent. But in severe features, eclampsia, or recurrent seizure prevention, magnesium is core treatment.

Magnesium also teaches a broader lesson. A seizure in pre-eclampsia is not just epilepsy occurring during pregnancy. It is the clinical endpoint of cerebral involvement in a systemic endothelial disease. That is why magnesium is preferred over routine anticonvulsants for eclampsia, and why treating the BP and planning delivery remain part of seizure care.

Pulmonary Edema Prevention

The third emergency target is pulmonary edema prevention.

Pre-eclampsia is a capillary leak disease. The patient may be edematous and still have a vulnerable intravascular volume. Giving large volumes of crystalloid can push fluid into the lungs. NICE limits maintenance fluids in severe pre-eclampsia to about eighty milliliters per hour unless there are ongoing losses. ACOG also cautions against aggressive fluid administration.

So in severe disease, fluids are medication. They need an indication, a rate, and reassessment.

This is especially important around labor, epidural placement, cesarean delivery, oxytocin, hemorrhage, and oliguria. A low urine output in severe pre-eclampsia does not automatically mean the patient needs repeated fluid boluses. It may reflect renal vasospasm and endothelial injury. Before giving fluid, ask whether there is hemorrhage, whether the lungs are clear, what the oxygen saturation is, what the creatinine is doing, and whether the patient is already intravascularly vulnerable.

Organ Trajectory

The fourth target is organ trajectory.

One platelet count, one creatinine, and one liver enzyme result are a snapshot. The trend tells you whether the disease is accelerating. A platelet count of one hundred ten thousand and falling can be more concerning than a stable one hundred thirty thousand. A creatinine that doubles from a low pregnancy baseline is important even if the absolute number does not look dramatic outside pregnancy.

HELLP Syndrome

Now HELLP syndrome.

HELLP means hemolysis, elevated liver enzymes, and low platelets. ACOG criteria include LDH at least six hundred, AST or ALT at least twice normal, and platelets below one hundred thousand. It can present with right upper quadrant pain, epigastric pain, nausea, vomiting, or malaise. It can occur postpartum. It can appear without classic hypertension or proteinuria.

Steroids may temporarily improve platelet counts or labs, but SOMANZ and ISSHP do not recommend corticosteroids to treat HELLP itself because outcome benefit is not proven. Delivery and maternal stabilization are the real treatment framework.

Eclampsia

Now eclampsia.

Eclampsia is not always the final step in a neat progression from mild disease to severe disease to seizure. ACOG emphasizes that seizures can occur without classic preceding hypertension or proteinuria, and many patients have warning symptoms such as headache, visual symptoms, epigastric pain, or altered mental status. The differential diagnosis includes epilepsy, intracranial bleeding, posterior reversible encephalopathy syndrome, reversible cerebral vasoconstriction syndrome, thrombotic thrombocytopenic purpura, hemolytic uremic syndrome, and other neurologic or hematologic emergencies. But in a pregnant or postpartum patient, eclampsia must stay high on the list until it is safely excluded.

When a seizure occurs, the first move is maternal stabilization. Call for help. Protect the patient from injury. Position her laterally. Support airway and oxygenation. Suction if needed. Give magnesium. Treat severe hypertension. Then reassess the fetus.

This sequence matters.

Gabbe emphasizes that fetal bradycardia and uterine hyperactivity can occur during and after a seizure and often resolve after maternal stabilization. A bad tracing during the seizure does not mean the first move is immediate cesarean. The mother is the resuscitation target. Stabilize her, then decide delivery route and timing.

Gestational-Age Decision

Now the gestational-age decision.

At thirty four weeks or beyond, ACOG recommends delivery for pre-eclampsia or gestational hypertension with severe features after maternal stabilization. Delivery should not be delayed just to complete late-preterm steroids.

Before thirty four weeks, expectant management may be considered only in carefully selected stable patients, in a center with maternal and neonatal intensive resources. The patient must remain stable. The fetus must remain reasonably stable. The team must be able to monitor closely and deliver quickly if the trajectory changes.

Expectant management is not "doing nothing." It is intensive risk management.

That means senior obstetric involvement, anesthesia awareness, neonatal counseling, clear criteria for abandoning expectant management, repeated maternal labs, close BP control, symptom checks that are actually taken seriously, and fetal surveillance appropriate to the gestational age and placental status. If the institution cannot provide that system, the patient is not a candidate for expectant management there, even if she looks stable at this moment.

Delivery is indicated at any gestational age for uncontrolled severe BP despite therapy, eclampsia, stroke, persistent severe neurologic symptoms, pulmonary edema,

worsening renal function, progressive thrombocytopenia, HELLP syndrome, DIC, abruption, nonreassuring fetal status, fetal demise, or severe fetal compromise.

Let us put this into a bedside loop.

Prevent stroke: treat severe BP.

Prevent seizure: give magnesium when indicated.

Avoid pulmonary edema: restrict fluids and monitor oxygenation and urine output.

Track organs: platelets, creatinine, liver enzymes, symptoms.

Track placenta: fetal tracing, growth, fluid, Doppler.

Decide whether time helps or harms.

That loop is the maternal safety system.

Resident Reflex

The resident reflex is this: severe features mean the disease has entered an organ-danger phase. The first questions are not academic. Is the brain protected from stroke? Is the patient protected from seizure? Are we avoiding pulmonary edema? Are the organs worsening? And does this pregnancy still benefit from time?